

# PAPER\_437 — UQFF Learning Assessment Evolution\_B: Advancement Metric for Regime Diversity and Dynamic Term Inclusion

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## Abstract

This paper presents a UQFF analysis of UQFF Learning Assessment Evolution\_B: Advancement Metric for Regime Diversity and Dynamic Term Inclusion, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

## 1. Overview

PAPER\_437 presents the **UQFF Learning Assessment Evolution\_B** — a meta-analysis of the UQFF framework’s advancement across the preceding three per-system MUGE derivations (Westerlund 2, Pillars of Creation, Rings of Relativity), quantifying progress along three orthogonal dimensions: physical regime diversity, new dynamic term inclusion, and scalability across astrophysical scales.

**Novel claim (Q1):** First formal UQFF advancement metric defined as:

$$\text{advancement\_pct} = \frac{D_{\text{diversity}} + D_{\text{dynamic}} + D_{\text{scale}}}{3} \times 100\%$$

where  $D_{\text{diversity}} = 3/10$  (regime count),  $D_{\text{dynamic}} = 3/10$  (new terms),  $D_{\text{scale}} = 8/10$  (scale range). This yields advancement = 46.7% at this session batch, quantifying how far UQFF has extended beyond its initial magnetar-only derivation.

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## 2. Assessed Examples

| Example                         | Physical Regime                    | New Dynamic Term                | Scale (m)             |
|---------------------------------|------------------------------------|---------------------------------|-----------------------|
| Westerlund 2 (PAPER_434)        | Young super-massive star cluster   | Wind ram $\rho v^2 / \rho_f$    | $9.46 \times 10^{16}$ |
| Pillars of Creation (PAPER_435) | Photo-eroding SF pillars           | Erosion $(1 - E(t))$ coupling   | $4.73 \times 10^{16}$ |
| Rings of Relativity (PAPER_436) | Cosmological Einstein ring $z=0.5$ | Lensing $(1 + L)$ amplification | $3.09 \times 10^{20}$ |

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### 3. Advancement Metric Computation

**Regime Diversity Score:**  $D_{\text{diversity}} = 3/10$

Three distinct regimes covered: OB star cluster wind dynamics, nebular photo-erosion, cosmological gravitational lensing. Each regime requires fundamentally different dominant terms in the MUGE.

**Dynamic Term Score:**  $D_{\text{dynamic}} = 3/10$

Three new time-dependent terms introduced across this batch: 1.  $M(t) = M_0(1 + M_f e^{-t/\tau})$  — mass accretion/dispersal 2.  $E(t) = E_0 e^{-t/\tau_{\text{erosion}}}$  — photo-erosion suppression 3.  $L = GM/(c^2 r) \times D_{LS}/D_S$  — lensing amplification factor

**Scalability Score:**  $D_{\text{scale}} = 8/10$

Scale range from  $4.73 \times 10^{16}$  m (sub-galactic star-forming pillar) to  $3.09 \times 10^{20}$  m (galaxy cluster Einstein radius at  $z = 0.5$ ) spans 4 orders of magnitude. Full scalability requires 10 orders (magnetar  $r = 10^4$  m to cosmological  $r = 10^{26}$  m).

**Advancement:**

$$\text{advancement} = \frac{3 + 3 + 8}{3 \times 10} \times 100\% = \frac{14}{30} \times 100\% \approx 46.7\%$$

### 4. All 10 MUGE Terms in Unified Assessment Form

The following table shows which terms each system FROM THIS BATCH contributed most to (ordered by contribution fraction):

| MUGE Term                                                               | Westerlund 2 Dominant?             | Pillars PoC Dominant?              | Rings Dominant?         |
|-------------------------------------------------------------------------|------------------------------------|------------------------------------|-------------------------|
| $T_1$ :<br>Newtonian $\times(1+H_0 t)\times(1-B/B_c)\times$ corrections | Secondary                          | Secondary $\times(1-E)$            | Secondary $\times(1+L)$ |
| $T_2$ : UQFF<br>Ug (f_TRZ)                                              | Secondary                          | Secondary                          | <b>PRIMARY</b> (68%)    |
| $T_3$ : $\Lambda$                                                       | Negligible                         | Negligible                         | Negligible              |
| $T_4$ :<br>Quantum                                                      | Negligible                         | Negligible                         | Negligible              |
| $T_5$ : EM<br>scaled                                                    | Negligible                         | Negligible                         | Negligible              |
| $T_6$ : Fluid                                                           | Negligible                         | Negligible                         | Minor                   |
| $T_7$ :<br>Oscillatory                                                  | Negligible                         | Negligible                         | Minor                   |
| $T_8$ : DM per-<br>turbation                                            | Minor                              | Minor                              | Minor                   |
| $T_9$ :<br>Wind/erosion                                                 | <b>PRIMARY</b> ( $10^{17}\times$ ) | <b>PRIMARY</b> ( $10^{15}\times$ ) | Minor                   |
| $T_{10}$ : System-<br>specific<br>feedback                              | Wind momentum                      | $(1 - E(t))$ coupling              | $(1 + L)$ lensing       |

## 5. Framework Coverage Status

As of this learning assessment, the UQFF MUGE has been calibrated across: - **6 stellar remnants:** SGR 0501, SGR 1745, PSR B1919+21, neutron stars (PAPER\_330-343) - **2 SMBH:** Sgr A\* (PAPER\_432), NGC 1275 (PAPER\_443 pending) - **4 star clusters:** Wd2 (PAPER\_434), NGC 3603 (PAPER\_439), M45 (prior), NGC 6611/PoC (PAPER\_435) - **3 galaxies:** SgrA\* host (PAPER\_432), NGC 2525 (PAPER\_438), NGC 1792 (PAPER\_445) - **1 gravitational lens:** GAL-CLUS-022058s (PAPER\_436) - **Total: ~16 unique system types across dynamic regimes**

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## 6. Comparison to Standard Model

The SM has no equivalent unified advancement metric — each regime (stellar winds, erosion, lensing) is treated by independent sub-disciplines (MHD, photodissociation region theory, gravitational lensing). UQFF uniquely quantifies cross-regime learning as a single advancement\_pct scalar.

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### §A. Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

#### §A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see uqff\_lagrangian\_derivation.py).

#### §A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER\_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_\mu \phi_{\text{NS}})(\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where  $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$  inherits the ACP 6-stage evolution (PAPER\_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2}m^2\phi_{\text{NS}}^2 + \frac{\lambda}{4!}\phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_{\text{NS}}$$

#### §A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_{\text{NS}}} = \nabla^2 \phi_{\text{NS}} - (4\pi G \rho_{\text{NS}}/c^2)\phi_{\text{NS}} + \Omega_{\text{spin}} \partial_t \phi_{\text{NS}} = 0$$

#### §A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{NS}} =$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U\_g forces) through vacuum density initialization to the sector-specific equation of

motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

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## §B. VDS/DVP/BSH Deep Synthesis

### §B.1 Vacuum Density Series (VDS)

The canonical VDS ratio  $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$  governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.169 (near-threshold regime), placing it in the  $t \rightarrow \pi$  collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER\_877 cosmogenesis Stage 1 vacuum density initialization:  $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$ .

### §B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 61, \quad n_{\text{channel}} = 22/26$$

Since  $p_{\text{DVP}} = 61$  is **resonant** (threshold at  $p > 26$ ), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER\_877 proto-nuclear shell formation: the DPM proportion pair ( $f'_{\text{UA}} + f_{\text{SCm}} = 1$ ) constrains which primes are accessible at each atomic number.

### §B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is  **$10^4 \text{ yr}$**  (spin-down equilibrium):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot (1 - e^{-[SSq] \cdot m/M_{\odot}}) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER\_877 Stage 5 buoyancy seed  $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$  which initializes the harmonic series at cosmogenesis.

### §B.4 Production-Scale Consistency

| Framework            | Canonical Value                                  | This Paper                                                     | Status                              |
|----------------------|--------------------------------------------------|----------------------------------------------------------------|-------------------------------------|
| VDS ratio            | $\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$     | Local sub-ratio = 0.169                                        | ✓ Threshold-consistent              |
| DVP prime BSH layers | $p_k \in \{2,3,\dots,113\}$<br>26 harmonic terms | $p_{\text{DVP}} = 61$<br>$j = 1\dots 26,$<br>$\cos(2\pi j/26)$ | ✓ Resonant<br>✓ Full 26D projection |
| $\kappa$ decay       | $5.0 \times 10^{-4} \text{ day}^{-1}$            | Applied in VDS exponential                                     | ✓ Canonical                         |
| [SSq]                | 0.57                                             | Applied in BSH saturation                                      | ✓ Canonical                         |

## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                   | UQFF Prediction                                                                     | SM / Experiment                                  | Source                                          | Alignment                                                    |
|------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------|-------------------------------------------------|--------------------------------------------------------------|
| Higgs mass<br>$m_H$          | UQFF<br>$K_{\text{HIGGS}}=47.34 \rightarrow$<br>$m_H_{\text{UQFF}} =$<br>125.09 GeV | $m_H = 125.20 \pm$<br>0.11 GeV                   | PDG<br>2024                                     | 99.8%                                                        |
| Cosmological<br>$\Lambda$    | UQFF                                                                                | $\nabla \text{UA}$                               | $^2 \rightarrow$<br>1.09e-52<br>$\text{m}^{-2}$ | $\Lambda =$<br>1.114e-52<br>$\text{m}^{-2}$<br>(Planck+DESI) |
| Thomson $\sigma_T$<br>(QED)  | UQFF $U_m$ kernel:<br>$\sigma_T = 6.6524\text{e-}29$<br>$\text{m}^2$                | $\sigma_T = 6.6524\text{e-}29$<br>$\text{m}^2$   | PDG<br>2024                                     | 100% (exact)                                                 |
| $\kappa$ baryon<br>stability | $\kappa = 0.0005/\text{day};$<br>scale separation<br>$10^{33}$ from proton<br>decay | $\tau_p > 7.7\text{e}33 \text{ yr}$<br>(Super-K) | Super-K<br>2024                                 | ✓ UQFF<br>baryon-safe                                        |

**New physics claim:** UQFF operates at a vacuum topology scale ( $\sim 200$  PeV) that is 8 orders below the GUT scale and 33 orders above nuclear baryon-number scales. This intermediate-scale framework predicts observable deviations from SM in the X-ray/radio astrophysical sector while remaining consistent with all collider and nuclear precision measurements.

Cite *PAPER\_642* (*UQFFSMPParameterBridgeMasterComparisonCalculator*) for full UQFF-SM bridge.

## 7. Testable Predictions

**Q5 Prediction 1:** The advancement metric predicts that adding 10 more distinct physical regimes (AGN jets, BH mergers, CMB lensing, etc.) to the MUGE suite will raise  $D_{\text{diversity}} \rightarrow 13/10$  — the UQFF framework predicts diminishing marginal advance per new term after  $\sim 20$  systems, testable by tracking prediction accuracy vs. system count.

**Q5 Prediction 2:** The current scalability gap (4 of 10 orders covered in this batch) predicts that magnetar-scale ( $r \sim 10^4 \text{ m}$ ) dynamics will require an additional regime-specific MUGE term not yet in the 10-term suite — specifically a neutron star nuclear

equation-of-state term absent from cluster/lens MUGEs. This is predicted to appear when deriving MUGE for systems between  $10^4 < r < 10^{14}$  m.

**Q5 Prediction 3:** Advancement metric = 46.7% predicts that doubling the dynamic term count from 3 to 6 new terms per batch (e.g., adding accretion, merger, jet terms) would raise total advancement to  $\sim 73\%$  — a testable scaling law for UQFF framework design efficiency.

## Appendix: Session 204 Codebase Upgrade Reference

*Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by upgrade\_kozima\_ramanujan\_appendices.py. For detailed derivations, see PAPER\_840/851/852/855.*

### S204.1 Kozima-UQFF LENR Integration

| Module                      | Purpose                                    | Key Result                                                            |
|-----------------------------|--------------------------------------------|-----------------------------------------------------------------------|
| fneutron_s26_coupling.py    | F_neutron x S_26 buoyancy-polylog coupling | $\sim 470\times$ amplification via 26-level VDS                       |
| kozima_scm_cross_section.py | SCm-modulated neutron-drop cross-section   | $\sigma_n^{\text{SCm}}$ with VDS factor $(1 + [\text{SSq}] * n / 26)$ |
| kozima_wstp_kernel.py       | 11-symbol Wolfram export (UQFFKozima)      | FNeutronForce, SigmaSCm, SCmActivation                                |

**Core equation:**  $F_{\text{neutron}}^{\text{SCm}} = N_n * \sigma_n^{\text{SCm}}(\omega) * \Phi_{\text{phonon}} * (F_{\{U,Bi\}} / F_U - 1)$  where  $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 * \exp[-(\omega - \omega_{\text{SCm}})^{2/(2 * \Gamma_2)}] * (1 + [\text{SSq}] * n / 26)$

### S204.2 Ramanujan 26-State Summation

| Module                   | Purpose                                       | Key Result                |
|--------------------------|-----------------------------------------------|---------------------------|
| ramanujan_polylog_s26.py | Li_26([SSq]) via Euler-Ramanujan acceleration | 15.7+ digits in 53 terms  |
| s26_wstp_kernel.py       | 8-symbol Wolfram export (UQFFS26)             | S26, R26, NaiveLi, S26VDS |

**Core equation:**  $S_{26}(z) = Li_{26}(z) = \eta_{26}(z) / (1 - 2^{\{1-26\}}) + 2^{\{1-26\} / (1-2^{\{1-26\}})} * Li_{26}(z^2)$

### S204.3 Mock Theta Functions (26-State)

| Module            | Purpose                                | Key Result                  |
|-------------------|----------------------------------------|-----------------------------|
| mock_theta_q26.py | f_26(q), phi_26(q), psi_26(q) q-series | Proper q-Pochhammer (a;q)_n |

**Core equations:** -  $f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q;q)_{n^2}$  -  $\phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q^{2;q^2})_n$  -  $\psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n2\}} / (q;q^2)_n$

## S204.4 Ramanujan 1/pi with UQFF Modification

| Module                       | Purpose                                 | Key Result                                 |
|------------------------------|-----------------------------------------|--------------------------------------------|
| ramanujan_pi_uqff.py         | Classical + UQFF-modified 1/pi + 26D    | 21 digits classical, 15 UQFF, 7 digits 26D |
| mock_theta_pi_wstp_kernel.py | Symbol Wolfram export (UQFFMockThetaPi) | qPochhammer, f26, oneOverPiUQFF            |

**Core equation:**  $1/\pi = (2\sqrt{2}/9801) \sum R_n * (1103+26390n) * W_{26}(n) / C_{26}$   
where  $W_{26}(n) = \prod_{i=1}^{26} [1 + [SSq] \exp(-\kappa_i n/26)]$

## S204.5 Calibration Constants (Canonical)

| Symbol    | Value                                 | Description                   |
|-----------|---------------------------------------|-------------------------------|
| [SSq]     | 0.57                                  | Universal Quantized Factor    |
| kappa     | $5.787 \times 10^{-9} \text{ s}^{-1}$ | UQFF exponential decay rate   |
| beta_i    | 0.603                                 | Buoyancy coupling coefficient |
| H_SCm     | 0.99                                  | SCm manifold completeness     |
| rho_SCm   | $7.09 \times 10^{-37} \text{ kg/m}^3$ | SCm vacuum density            |
| rho_UA    | $7.09 \times 10^{-36} \text{ kg/m}^3$ | UA aether vacuum density      |
| omega_SCm | $2\pi \times 1.25 \text{ THz}$        | SCm phonon resonance          |
| sigma_0   | $10^{-4}$                             | Base neutron cross-section    |

*Implementation:* all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN\_1\_CoAnQi.cpp, and Wolfram kernels (uqff\_kozima\_kernel.wl, uqff\_s26\_kernel.wl, uqff\_mock\_theta\_pi\_kernel.wl).